Effect of cooking on garlic (Allium sativum L.) antiplatelet activity and thiosulfinates content.

The raw form of garlic and some of its preparations are widely recognized as antiplatelet agents that may contribute to the prevention of cardiovascular disease. Herein, we examined the in-vitro antiaggregatory activity (IVAA) of human blood platelets induced by extracts of garlic samples that were previously heated (in the form of crushed versus uncrushed cloves) using different cooking methods and intensities. The concentrations of allicin and pyruvate, two predictors of antiplatelet strength, were also monitored. Oven-heating at 200 degrees C or immersing in boiling water for 3 min or less did not affect the ability of garlic to inhibit platelet aggregation (as compared to raw garlic), whereas heating for 6 min completely suppressed IVAA in uncrushed, but not in previously crushed, samples. The latter samples had reduced, yet significant, antiplatelet activity. Prolonged incubation (more than 10 min) at these temperatures completely suppressed IVAA. Microwaved garlic had no effect on platelet aggregation. However, increasing the concentration of garlic juice in the aggregation reaction had a positive IVAA dose response in crushed, but not in uncrushed, microwaved samples. The addition of raw garlic juice to microwaved uncrushed garlic restored a full complement of antiplatelet activity that was completely lost without the garlic addition. Garlic-induced IVAA was always associated with allicin and pyruvate levels. Our results suggest that (1) allicin and thiosulfinates are responsible for the IVAA response, (2) crushing garlic before moderate cooking can reduce the loss of activity, and (3) the partial loss of antithrombotic effect in crushed-cooked garlic may be compensated by increasing the amount consumed.